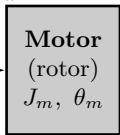


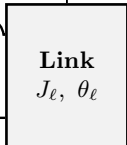
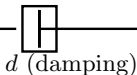
KUKA iiwa14 (7 axes)
meshes / masses / inertias
from MuJoCo Menagerie

exploded view

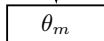
actuator
torque τ_m



K (gear stiffness)



next link



Motor-side encoder
(what the controller
assumes)

**Joint-side / out-
put encoder $\rightarrow$ ILC**
(reads the true link angle; true
TCP is used for evaluation only)

$$\theta_\ell = \theta_m + \Delta$$

$$\Delta = \theta_m - \theta_\ell = \text{transmission error}$$

τ_d (link disturbances, unknown to the controller) =
gear transmission error $K\varepsilon(\theta)$ + Stribeck friction +
backlash + thermal drift $K k_T T(t)$